

IoT Security Assignment - Lab Assignment No. 05

Project Overview

This repository contains the practical solution for Lab Assignment No. 05, which focuses on simulating security aspects of Internet of Things (IoT) devices and Cyber-Physical Systems (CPS).

The assignment was implemented using the Python programming language and covers the first two practical parts of the task.

Implemented Parts

Part	Title	Summary
Part I	IoT Device Data Encryption Simulation	Simulates the generation of random sensor data (temperature and humidity) and its encryption using the AES in GCM mode (Authenticated Encryption) algorithm to ensure confidentiality and integrity during transmission.
Part II	IoT Device Security Lifecycle Simulation	A conceptual simulation of the five essential security lifecycle stages for embedded/IoT devices, with event logging and timestamps.

Requirements and Installation

Running the code requires a Python 3.x environment and the [pycryptodome](#) cryptography library.

Prerequisites

- Python 3.x

Installation

To install the required cryptography library, use the following command:

```
pip install pycryptodome
```

How to Run

Assuming the code is saved in a single file named `iot_security_assignment.py`, you can run the simulation for both parts using the following command:

```
python iot_security_assignment.py
```

Simulation Results

Part I: IoT Device Data Encryption Simulation

This part demonstrates the secure encryption and decryption process for data between an IoT sensor (IoT Device) and a base station (Server/Base Station). **AES-GCM** was used to ensure:

- 1 **Confidentiality:** Through encryption.
- 2 **Integrity and Authenticity:** Through the Authentication Tag.

Example Output (Extracted from Screenshots):

```
--- Part I: IoT Device Data Encryption Simulation ---

Secret Key (16 bytes): [Base64 Encoded Secret Key]

[IoT Device] Raw Sensor Data (Plaintext):
{"device_id": "IoT-Sensor-001", "timestamp": "2025-11-29T10:00:00Z",
"temperature_c": 35.24, "humidity_percent": 53.4}

[Transmission] Encrypted Data Components (Ciphertext, Nonce, Tag):
Nonce (IV): [Base64 Encoded Nonce]
Ciphertext: [Base64 Encoded Ciphertext]
Authentication Tag: [Base64 Encoded Authentication Tag]

[Server] Decrypted Sensor Data:
{"device_id": "IoT-Sensor-001", "timestamp": "2025-11-29T10:00:00Z",
"temperature_c": 35.24, "humidity_percent": 53.4}

[Result] SUCCESS: Decrypted data matches the original raw data.
```

Part II: IoT Device Security Lifecycle Simulation

This part simulates the five stages of the IoT/embedded device security lifecycle, logging each stage with a timestamp.

The Five Simulated Stages:

- 3 **Threat Modeling:** Identifying potential risks.
- 4 **Secure Boot Initialization:** Ensuring firmware integrity.
- 5 **Secure Key Injection:** Simulating the secure introduction of cryptographic keys.
- 6 **OTA Firmware Update Check:** Verifying the device is running the latest secure firmware.
- 7 **Secure Decommissioning:** Securely wiping secrets and sensitive data before disposal.

Example Output (Extracted from Screenshots):

```
[2025-11-30 10:30:59.791549] [Stage 1] Threat model created (assets, attack vectors, mitigations).
```

```
[2025-11-30 10:31:00.791845] [Stage 2] Secure boot verified – firmware integrity OK.  
[2025-11-30 10:31:01.792138] [Stage 3] Keys injected securely (Mock Key: AB12-CD34-EF56).  
[2025-11-30 10:31:02.792462] [Stage 4] OTA update verified – device running latest secure firmware.
```

```
[2025-11-30 10:31:03.792728] [Stage 5] Device decommissioned – all secrets wiped securely.
```